Avery (Zixuan) Yu

Seattle, WA zixuan.jhu@gmail.com 909-292-3960 LinkedIn

Education

University of Washington *Sept 2024 - May 2028*
PhD in Inforamtics: Biomedical and Health Informatics | GPA: 3.9/4.0

- Top Scholar Award
- Data Science Specializtion

University of Colorado, Boulder *Mar 2024 - May 2027*
Master of Science: Computer Science | GPA: 4.0/4.0

Johns Hopkins University Bloomberg School of Public Health *Aug 2021 - May 2023*
Master of Health Science: Genetic Epidemiology | GPA: 3.9/4.0

Zhejiang University City College *Aug 2016 - May 2021*
Bachelor of Medicine, Bachelor of Surgery - MD equivalent | GPA: 3.9/4.0

Publications

Evaluating Bias in Machine Learning Algorithms: Screening for Inpatient Behavioral Codes
Chipkin, J.*, **Yu, A.***, Dobbins, N. et al.
Academy of Consultation-Liaison Psychiatry, 2025 https://doi.org/10.1016/j.jaclp.2025.10.223

Identity by descent and local ancestry mapping of HCV spontaneous clearance in populations of diverse ancestries
Yu, Z., Abdel-Azim, S., Duggal, P., & Vergara, C., *BMC Genomics* 26, 661 (2025)
https://doi.org/10.1186/s12864-024-11076-6

WoundcareVQA: A Multilingual Visual Question Answering Benchmark Dataset for Wound Care
Yim, W., Abacha, A., Chen, C., Xu, J., Subbarao, A., Doerning, R., **Yu, Z.**, et al.,
Journal of Biomedical Informatics 170, 104888. (2025) https://doi.org/10.1016/j.jbi.2025.104888

Traj-CoA: Patient Trajectory Modeling via Chain-of-Agents for Lung Cancer Risk Prediction
Zeng, S., Fu, Y., Zhou, S., **Yu, Z.**, et al., *NeurIPS* GenAI4Health Workshop (2025)

Experience

Research Engineer Intern *June 2025 - Sept 2025*
Medeloop, Inc

- **Semantic Search Benchmarking:** Designed and executed a benchmarking pipeline for medical code retrieval (CPT, ICD-10, HCPCS) from user-entered terms, comparing Medeloop outputs with OpenAI GPT and Meta Llama baselines against two gold standards: (i) customer-provided sets and (ii) internally curated, publication-derived value sets.
- **RAG-Enhanced Plotting Pipeline**: Deployed a retrieval-augmented generation (RAG) module into the multi-agent framework to automatically generate scientifically guided, publication-quality plots —boosting pipeline completion rates and improving visualization aesthetics.

Graduate Research Assistant *Seattle, WA*
University of Washington *Sept 2024 - present*

- **Deployed AI Pipelines:** Designed and deployed a scalable pipeline to process and summarize 3,000+ clinical notes using Llama and leveraged expert annotations.
- **LLM-Driven Predictive Modeling:** Trained BERT-based models to classify clinical notes, achieving 0.81 F1 in predicting violence against healthcare providers.
- **Guideline Development for Annotation:** Developed structured guidelines for medical concept identification and AI-generated summary evaluation, supporting annotation.

Bioinformatics Research Analyst *St. Louis, MO*
Washington University in St. Louis, School of Medicine *June 2023 - June 2024*

- Designed and implemented an ETL pipeline for genome-wide association studies (GWAS), integrating dosage data processing, QC filtering, and parallelized computations, reducing 70% of the processing time.
- Maintained and optimized genomic and phenotype databases, ensuring efficient storage, retrieval, and integration of multi-source datasets.

Graduate Research Assistant *Baltimore, MD*
Johns Hopkins University *Oct 2021 – May 2023*

- Conducted genetic studies focusing on host susceptibility to HCV clearance, analyzing rare variants in multi-ethnic cohorts.
- Conducted IBD haplotype mapping on 1,869 individuals of African ancestry and 1,739 individuals of European ancestry. Detected over 7.3 million combined IBD segments.
- Identified a genomic signals in European populations near the MHC region, contributing to understanding immune response.

Graduate Student Data Analyst *Remote*
Johns Hopkins University, Johns Hopkins Center for Communication Programs *Sept 2022 - April 2023*

- Developed and deployed statistical models to assess questionnaire data quality, internal reliability, and construct validity, ensuring robust survey methodologies and actionable insights.
- Optimized data workflows for scalability and performance, leveraging R, ShinyR, and Python for efficient data processing and visualization.

Student Physician *Hangzhou, China*
Sir Run Run Shaw Hospital - Affiliated Hospital of Zhejiang University *June 2020 - June 2021*

- Rotated through multiple medical departments, assisting senior physicians in patient care.
- Documented and maintained accurate EHRs, supporting effective diagnosis and treatment.

Relevant Coursework

Informatics Research Methods, Clinical Data Analysis, Data Science I-II, Statistical Machine Learning, Deep Learning and Medical Imaging, Statistical Computing, Multilevel Statistical Models, Methods in Biostatistics Series, Data Structure

Skills

Languages: Python, Java, R, SQL, HTML/CSS, Shell, Bash

Technologies: PyTorch, Flask, REST APIs, Git, Docker, LaTeX, DICOM, AWS (S3, EC2), ShinyR, Git, Transformers, LaTex, Shell Scripting, NumPy, Matplotlib